



INTRODUCTION TO BUILDING MATERIALS - TIMBER

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Introduction



Timber: Refers to wood that is specifically chosen for use in engineering projects, prized for its combination of durability and visual appeal.



Wood: The natural, fibrous material harvested from trees, serving as the raw resource for various applications.



Lumber: Processed wood cut into boards or planks, primarily utilized in building and construction tasks.





Introduction

- ❖ The terms "wood" and "timber" are often used interchangeably, but they actually refer to different things.
- ❖ "Wood" encompasses all forms of wood, whether it's used for burning, crafting furniture, or building structures.
- ❖ When wood is specifically prepared and suitable for construction purposes, it is known as "timber."
- ❖ In essence, "wood" usually describes the raw, natural material as it comes from trees.
- ❖ "Timber," on the other hand, refers to wood that has been processed or treated by humans for practical use.
- ❖ So, you might gather wood from the forest to start a fire, but you'd visit a hardware store to buy timber for home improvement projects.



Introduction

- ❖ Timber is derived from trees and specifically refers to wood used for structural purposes.
- ❖ A living tree standing upright is called standing timber.
- ❖ **Rough Timber:** This is the stage when a tree has been felled and its trunk and branches are roughly cut into pieces of manageable lengths.
- ❖ **Converted Timber:** When rough timber undergoes further processing—being sawn and shaped into commercially usable sizes such as planks, logs, battens, posts, and beams—it is known as converted timber.



Classification of Trees

According to the mode of growth the trees are classified as :

➤ **EXOGENOUS TREES:**

which comprise the great majority of modern trees (all conifers, and all broadleaf trees), grow by the addition of new wood outwards, immediately under the bark.

➤ **ENDOGENOUS TREES:**

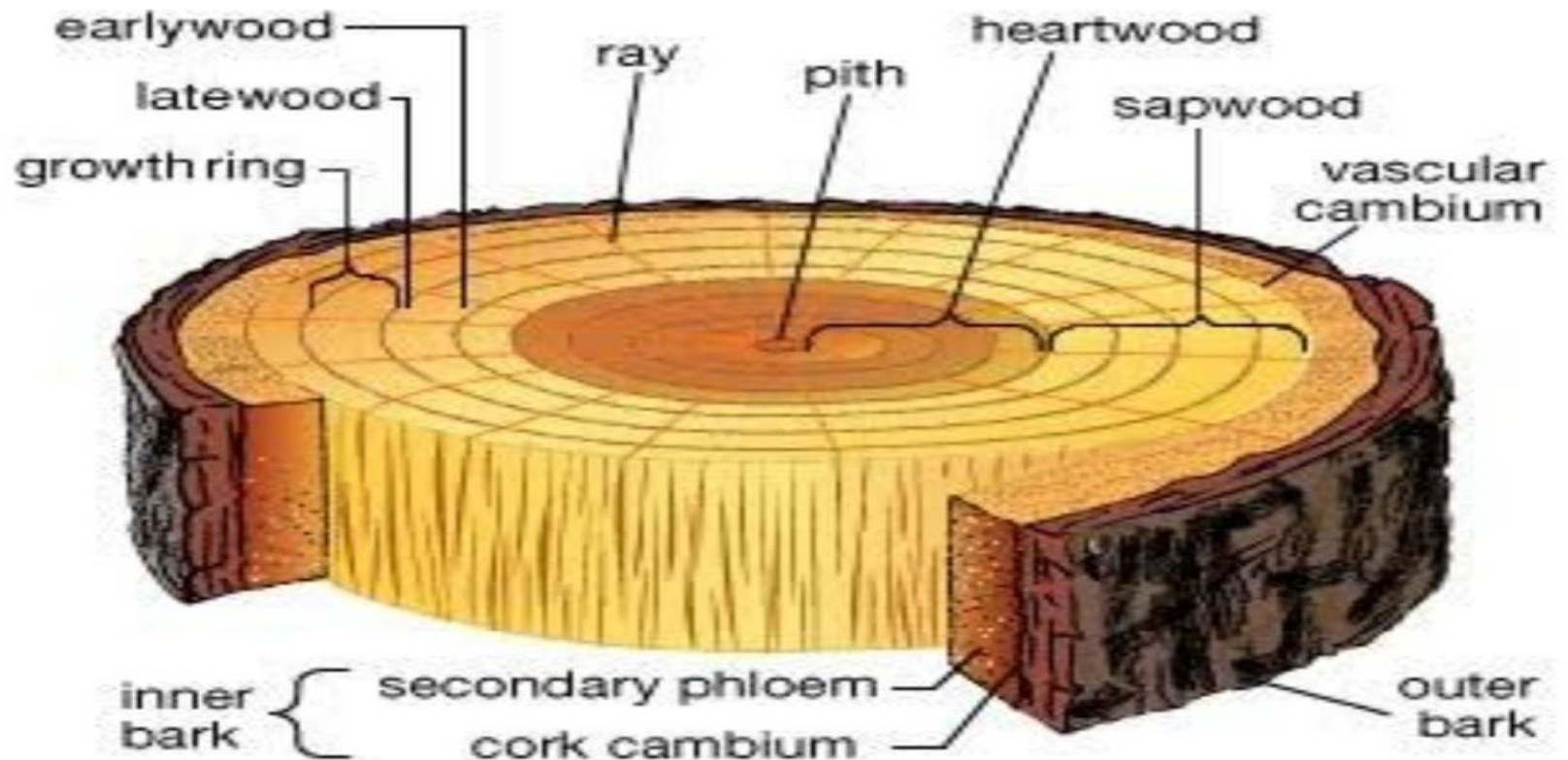
Grow by addition of new material inwards.(bamboo.cane,palm)



Structure of Exogenous Tree

Fig. a cross section of an exogenous tree in which names of all the parts have been given the different parts that can be distinguished are as follows:

- 1 Pith
- 2 Heart wood
- 3 Sapwood
- 4 Cambium layers
- 5 Inner bark
- 6 Outer bark
- 7 Medullary ray
- 8 Annual rings





Softwood v/s Hardwood

SOFTWOOD

- ❖ Softwood is not necessarily a softer material (less dense).
- ❖ Softwood comes from a conifer, which usually remains evergreen.
- ❖ Light- coloured.
- ❖ Growth rings distinct.
- ❖ Usually light in weight.
- ❖ Can be easily split

HARDWOOD

- ❖ Hardwood is not necessarily a harder material (more dense)
- ❖ Hardwood comes from a deciduous tree which loses its leaves annually.
- ❖ Generally dark- coloured.
- ❖ Growth rings are not as distinct as in softwood.
- ❖ Usually heavier in weight & Cannot be easily split



Felling

Felling:- Felling is the process of cutting down individual trees, an element of the task of logging. The person cutting the trees is a feller.

Methods

- ❖ **Hand felling** :-In hand felling, an axe, saw, or chainsaw is used to fell a tree, followed up by limbing, bucking in traditional applications. In the modern commercial logging industry, felling is typically followed by limbing and skidding.
- ❖ **Feller-Buncher:-** A feller- buncher is a motorized vehicle with an attachment which rapidly cuts and gathers several trees in the process of felling them. In cut-to-length logging a harvester performs the tasks of a feller-buncher additionally doing the delimbing and bucking of the trees as well.





Properties of Timber

1. Color
2. Appearance
3. Hardness
4. Specific Gravity
5. Moisture Content
6. Grain
7. Shrinkage and Swelling
8. Strength
9. Density
10. Toughness
11. Elasticity
12. Warping
13. Durability
14. Deffectless
15. Workability
16. Soundness
17. Free of abrasion

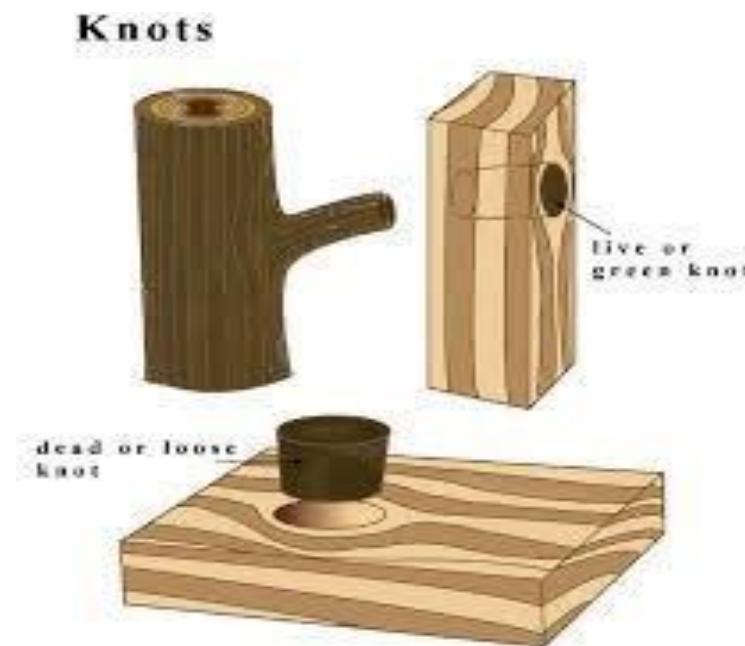


Defects in Timber

- ❖ **Knots**: Knots are the most common defects caused due to natural forces. During the growth of a tree, branches close to the ground or lower branches die. Bases of those branches remain in the tree as the trees grow. These bases may create imperfection known as knots.

Types of Knots: Knots are of two types.

- ❖ **Dead knots**: The remains of damaged branches after drying out they become loose and fall out.
- ❖ **Live knots**: They are sound and firm. If small, are not great of a defect.
- ❖ Live knots are usually not a problem as they remain firmly attached to the timber. But in dead knots, they are loosely attached and reduce strength. Knots decrease the strength of the wood and thus lower its value for structural uses. Knots cause serious defects when the load is perpendicular to the grains.





Defects in Timber

Twist: Twist in timber rotates the ends of the timber in opposite directions. The main reason behind this defect is twisting of the trees by the strong wind.



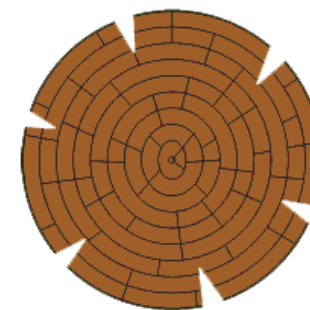


Defects in Timber

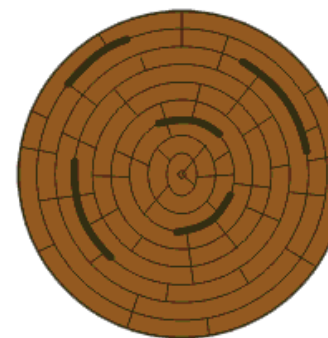
Shakes: Shakes are timber defects that occur around the annual ring or growth ring of a timber. In other words, cracks or splits in the woods are called shakes.

Types of shakes: Shakes can be classified into three main categories:-

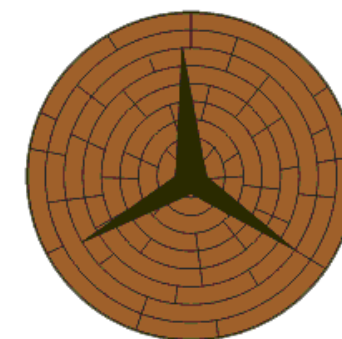
- 1. Star Shakes:** This type of shake starts propagating from the bark towards the sapwood and sometimes even towards the heartwood along the lines of medullary rays. Cracks are wider on the outer edge or bark and narrower on the inside (usually sapwood, sometimes heartwood). The main reasons behind star shakes are extreme heat or frosting during the growth of the trees and rapid or uneven seasoning after cutting off the timber. Extreme heat or frost causes temperature difference, which causes shrinkage leading to the crack.
- 2. Cup and/or Ring Shakes:** Cup shakes follow the annual growth ring. It is capable to separate the growth ring partially or completely. When the crack separates the annual ring completely, it is called ring shakes. So, all ring shakes are cup shakes, but all cup shakes are not a ring shape. Excessive frost action is the main reason for this type of crack.
- 3. Heart Shakes:** Unlike star shakes, heart shakes starts propagating from the pith to the sapwood along the lines of medullary rays. Shrinkage of the interior part of the timber causes this crack.



Star Shake



Ring Shake



Heart Shake



Defects in Timber

Rind Galls: The meaning of rind is bark and gall is abnormal growth. So abnormal growth of the bark of the trees is called rind galls. Improper cutting of branches causes this abnormal growth. Wood from this portion of the timber lacks strength and desirable in structure.



Upsets: Upsets in various wood indicate that the tree was subjected to crushing or compression. Improper felling of trees, heavy wind blowing during the young age of the tree these are the main reasons behind this type of defect.



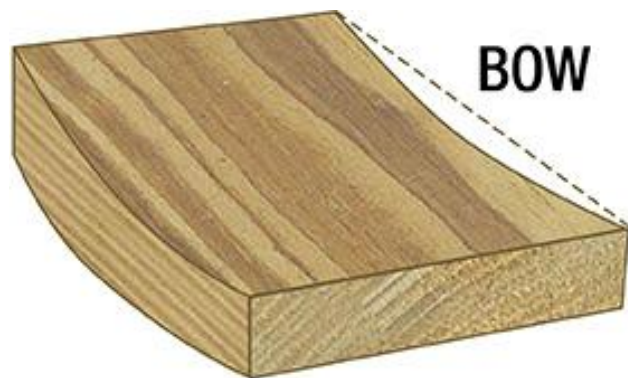


Defects in Timber

Defect in Timber due to Defective Seasoning

Faulty method of seasoning causes serious defects in woods. During seasoning of timber, exterior or surface layer of the timber dries before the interior surface. So, stress is developed due to the difference in shrinkage. In a perfect seasoning process, stress is kept minimum by controlling the shrinkage. Some of the defects resulting from defective seasoning are as follows:-

- 1.Bow:** Curvature formed in direction of the length of the timber is called bow.
- 2.Cup:** Curvature formed in the transverse direction of the timber is called a cup.
- 3.Check:** Check is a kind of crack that separates fibers, but it doesn't extend from one end to another.
- 4.Split:** Split is a special type of check that extends from one end to another.
- 5.Honey Combing:** Stress is developed in the heartwood during the drying process or seasoning. For these stresses, cracks are created in the form of a honeycomb texture.



BOW



CUP



CROOK



KNOT



SPLIT



TWIST



CHECK



SHAKE



WANE



Defects in Timber

Defects of Timber due to Defective Conversion

1. **Boxed Heart:** This term is applied to the timber, which is sawn in a way that the pith or the center heart falls entirely within the surface throughout its length.
2. **Machine Burnt:** Overheating is the main reason for this defect.
3. **Machine Notches:** defective holding and pulling causes this defect.
4. **Miscut:** erroneous cutting or sawing of wood causes this defect. Lack of experience in sawing and carelessness is the main reason for erroneous cutting.
5. **Imperfect Grain:** Mismatch in grain alignment.



Seasoning of Timber

- ❖ Air drying timber stack. Wood drying (also seasoning lumber or wood seasoning) reduces the moisture content of wood before its use when the drying is done in a kiln . The product is known as kiln dried timber or lumber whereas air drying is the more traditional method.

Methods of seasoning:-

- NATURAL DRYING
 - KILN DRYING
- ❖ There are two main ways of seasoning timber, Natural (Air) and Artificial (Kiln) drying. Both methods require the timber be stacked and separated to allow the full circulation flow of air, etc. around the stack. Air seasoning is the method used with the timber stacked in the open air



Seasoning of Timber

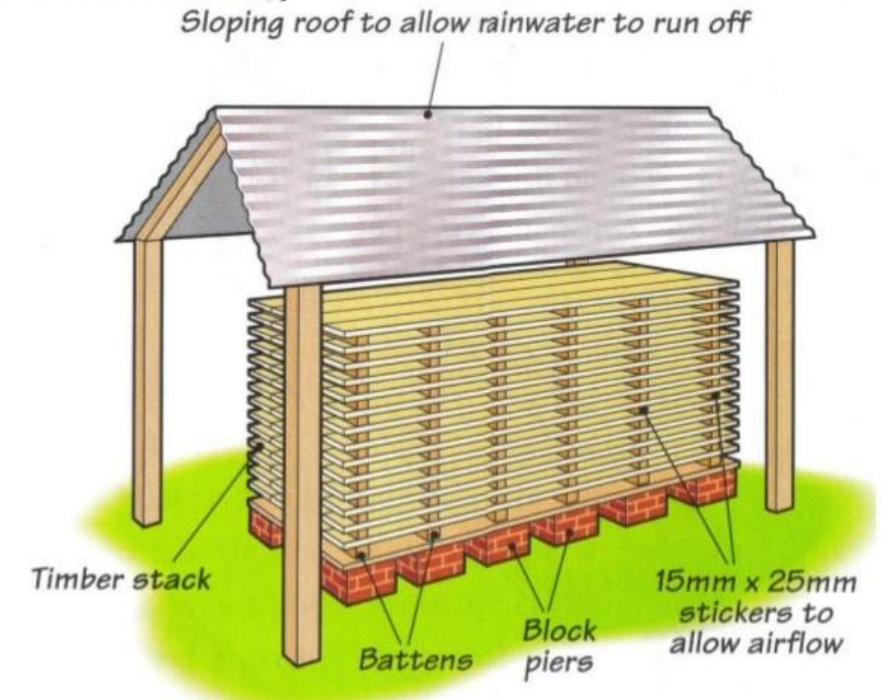




Air Seasoning

- ❖ In air seasoning of timber, logs of timber logs are stacked in layers in open air.
- ❖ Sufficient gap is ensured between the logs to ensure continuous ventilation.
- ❖ The logs are place on top of a platform which is 30cm above the ground.
- ❖ Ventilation of air through the logs facilitates sufficient reduction of water content in timber.
- ❖ Air seasoning of timber is a slow process of seasoning of timber.

Air Seasoning





Merits & Demerits of Air Seasoning

Advantages

- ✓ Low cost and requires simple equipment.
- ✓ Requires no fuel or electricity.
- ✓ Suitable for large-sized timber.
- ✓ Environment-friendly and easy to carry out.

Disadvantages

- Very slow and may take several months.
- Depends on weather conditions.
- Requires large storage space.
- Risk of defects such as warping, checking and splitting.



Klin Seasoning

- ❖ In this method the wooden pieces are kept on huge trollies and then it is placed inside the hot chamber of a furnace known as **kiln**.
- ❖ In this way wooden pieces are kept in a controlled atmospheric conditions for two weeks.
- ❖ The time depends upon the type of wood and initial moisture content. Blowers are also used.
- ❖ It is a quick drying method, and the moisture is reduced to 10%. Since seasoning is done at a lesser time through artificial means so the quality of wood is inferior in comparison to seasoned wood through natural seasoning.



Klin Seasoning

There are two main types of kiln used in artificial seasoning:

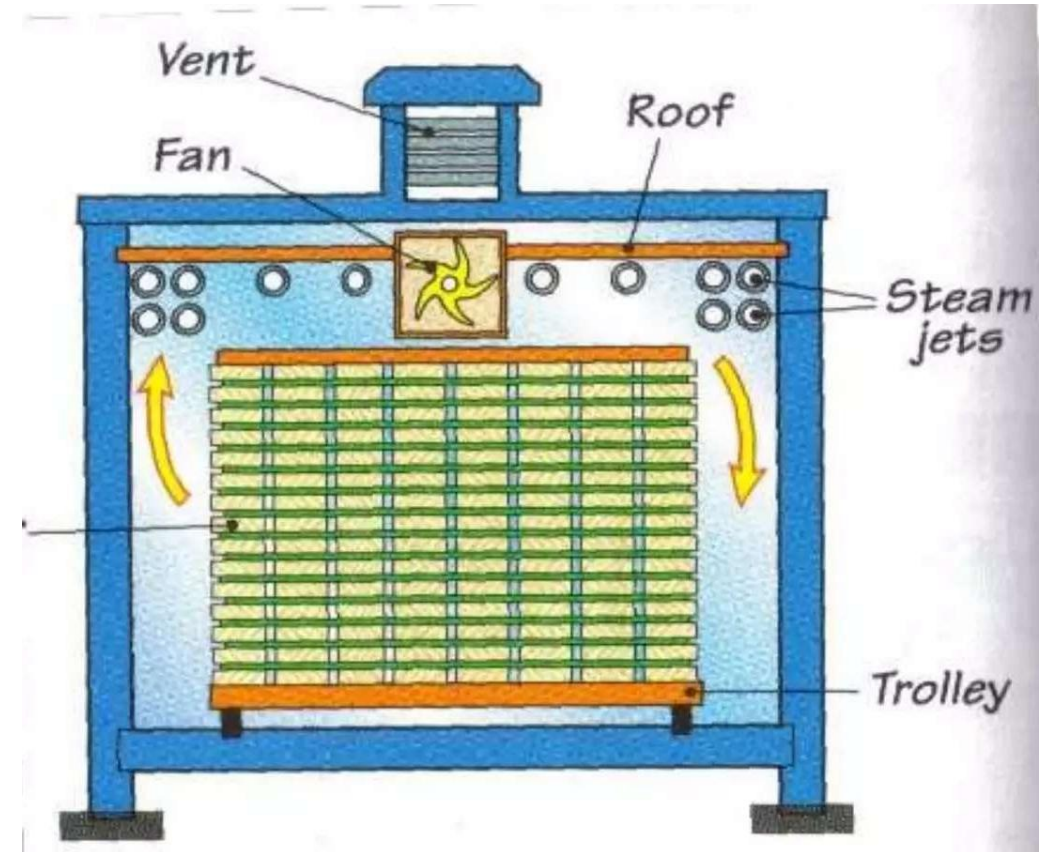
- Compartmental Kilns
 - Progressive Kilns
- ❖ Both methods rely on a controlled environment to dry out the timber and require the following factors:
- ❖ Forced air circulation by using large fans, blowers, etc.
 - ❖ Heat of some form provided by piped steam.
 - ❖ Humidity control provided by steam jets.
 - ❖ The amount and duration of air, heat and humidity depends on species, size, quantity, etc. In general, the atmosphere in the kiln at first will be cool and moist. The temperature is gradually increased and the humidity reduced until the required moisture content is achieved.



Klin Seasoning

Compartmental Kilns

- ❖ This kiln is a single enclosure container or building, etc.
- ❖ The timber is stacked same manner as air seasoning.
- ❖ Whole stack is seasoned using a programme of settings (temperature and humidity) until the whole stack is reduced to the MC required.

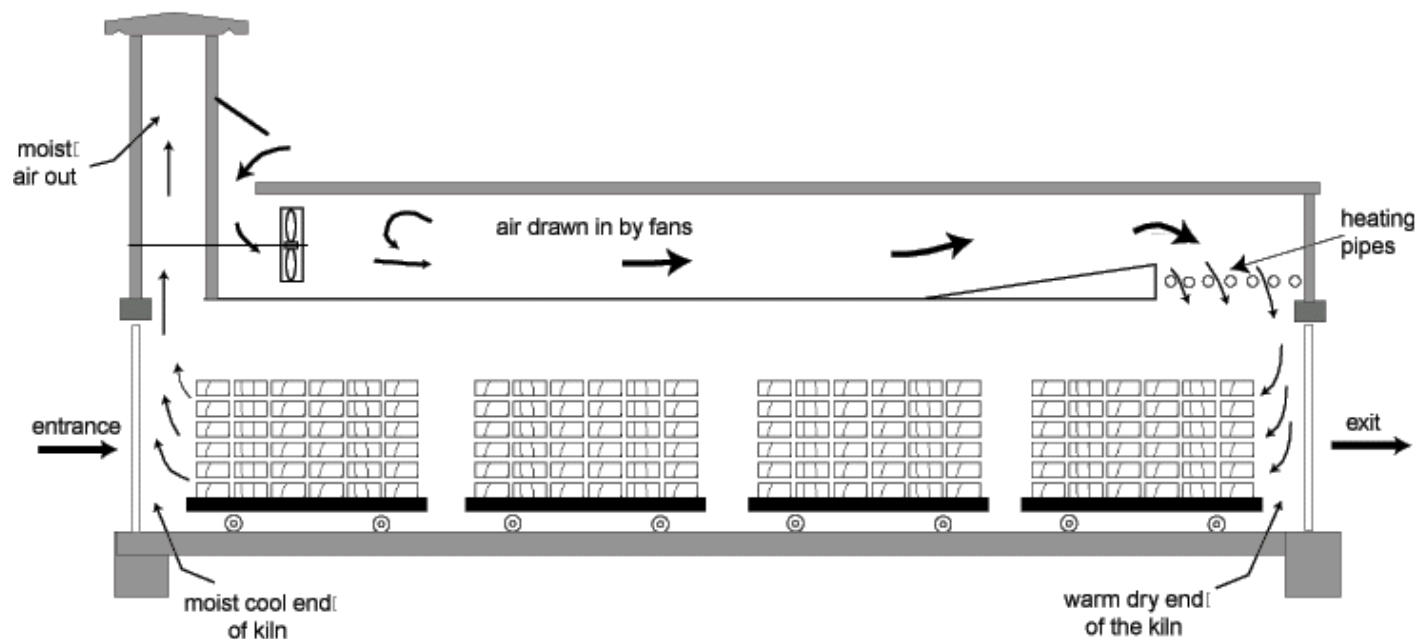




Klin Seasoning

Progressive Kilns

- ❖ A progressive kiln has the stack on trolleys that 'progressively' travel through a sequence of chambers.
- ❖ Each chamber has varying atmospheres that change the MC of the timber stack as it travels through.
- ❖ **Advantages of this system** – has a continuous flow of seasoned timber coming offline.





Merits & Demerits of Klin Seasoning

Advantages

- ✓ It dries quickly.
- ✓ It can be controlled.
- ✓ Achieve a lower moisture content.
- ✓ Defects associated with drying can be controlled.

✓ Disadvantages

- It is an expensive method.
- Requires supervision by a skilled operator.
- Uses a lot of energy.



Advantages of Timber

- ❖ Excellent insulator: houses they are built out of bricks also known to have outstanding insulating properties, but timber manages them
- ❖ Timber is an elastic material: At the opposite of brick or concrete walls, timber walls don't need to be very thick.
- ❖ Easy and fast to build: a timber construction can be build very easy and fast in contrast to a common brick or concrete construction
- ❖ Sustainable: timber is a renewable resource
- ❖ Reusable: timber can be used more than once. It is easy to recycle
- ❖ Price: to build a timber house or a timber construction is cheaper than any other common procedure which has the same requirements or sizes.
- ❖ Weight: timber is very light
- ❖ Strength: Timber framings are strong in real structural terms



Qualities of Good Timber

Good timber should have the following qualities

HARDNESS

A good quality timber should be hard enough to resist deterioration.

STRENGTH

It should have sufficient strength to resist heavy structural loads.

TOUGHNESS

It should have enough toughness to resist shocks due to vibrations. It should not break in bending and should resist splitting. Timbers having narrow annual rings, are generally the strongest.

ELASTICITY

It should have the property of elasticity so as to regain its original shape after removal of loads. This is a very important property to be considered if the timber is used in making sport goods.

DURABILITY

It should be able to resist attacks of fungi and worms and also atmospheric effects for a longer period of time.

DEFECTS

Timber should be prepared from the heart of a sound tree and be free from sap, dead knots, shakes and other similar defects.



Qualities of Good Timber

FIBRES AND STRUCTURE

It should have straight and closed fibers and compact medullary rays. It should give a clear ringing sound when struck. Dull heavy sound is an indication of internal decay. Its annual rings should be uniform in shape and colour.

structure of a timber

APPEARANCE AND COLOUR

Freshly cut surface should give sweet smell and present shining surface. It should have dark colour, as light colored timbers are generally weak in strength.

SHAPE AND WEIGHT

It should retain its shape during the process of seasoning. Heavy timbers are always stronger than light weight timbers.

WORKABILITY

It should be well seasoned and easily workable. Teeth of saw should not get clogged during the process of sawing. It should provide smoothened surface easily.



Uses of Timber

1. For heavy construction works like columns, trusses, piles.
2. For light construction works like doors, windows, flooring and roofing.
3. For other permanent works like for railway sleepers, fencing poles, electric poles and gates.
4. For temporary works in construction like scaffolding, centering, shoring and strutting, packing of materials.
5. For decorative works like showcases and furniture's.
6. For body works of buses, lorries, trains and boats
7. For industrial uses like pulps (used in making papers), card boards, wall papers
8. For making sports goods and musical instruments.



Preservation of Timber

The preservation of wood includes painting and other applied finishes. Timber preservation became necessary because naturally durable timbers became scarce, so ways of improving the durability of cheaper timbers were developed.

Timber needs to be protected from:

- Insects and fungi
- Wear and tear
- Effects of weathering



Preservation of Timber

The main forms of preservative are:

- Tar oil preservatives
- Water borne preservatives
- Solvent-based preservatives.

These preservatives can protect the timber in two ways.

1. They give physical protection by stopping fungi and insects getting at the wood.
2. They give chemical protection, making the wood poisonous to insects and fungi. Paint, for example, gives excellent physical protection.



Preservation of Timber

Coal Tar

Timber surface is coated with hot coal tar (because coal tar becomes workable when heated) with the help of a brush, and this process is known as tarring. Coal tar is fire-resistant.

Coal tar has an unpleasant smell and appearance and it makes timber unsuitable for painting.

Therefore, tarring is adopted for frames of doors and windows, rough timber work, etc. and is found to be most useful for parts embedded in the ground because of its cheapness and effective resistance.



Preservation of Timber

ASCU

ASCU is a special preservative which is developed at the Forest Research Institute, Dehradun.

Its composition is as below

1 part by weight of hydrated arsenic pentoxide,

3 parts by weight of blue vitriol or copper sulphate,

4 parts by weight of potassium dichromate, or sodium dichromate

These materials are available in powder form.

For the preparation of the solution, six parts by weight of ASCU are mixed in 100 parts by weight of water. The solution is then sprayed or applied on the timber surface. This preservative gives timber protection against the attack of white ants. Surfaces treated with this preservative can be painted, polished, varnished or waxed. This solution is odourless.



Preservation of Timber

Oil Paints

Oil paints work well on well-seasoned wood. They are typically applied in two or three coats. Oil paints protect the wood from moisture. Oil paints, if not properly seasoned, can cause timber decay by confining sap.

Solignum Paints

Solignum paints are applied with a brush while still hot. They are ideal for protecting wood from white ants. Solignum paints can be used to enhance the appearance of the timber by incorporating colour pigments.



Preservation of Timber

Creosote Oil

Tar is refined into creosote oil through distillation. It is either brown or black in colour. It has an awful odour. It is utilised in a unique way. First off, the wood has been properly dried and seasoned. It is then put into an airtight chamber, and the air inside is pumped out. Creosote oil is finally injected into the chamber at a high pressure of 0.7 to 1 N/mm² and 50°C. The timber is removed from the chamber after it has been allowed to soak up the creosote oil for two hours. Because creosote oil is flammable, it is not used for fireplace log work. Typically, it is used for things like railroad sleepers, poles, and wood piles.



Methods of Preservation of Timber

- ❖ Brushing of Timber Preservatives
- ❖ Spraying of Timber Preservatives
- ❖ Preservative Injecting Under Pressure
- ❖ Dipping and Steeping Method of Timber Preservation
- ❖ Charring Method of Timber Preservation
- ❖ Hot and Cold Open Tank Treatment of Timber



Merits of Preservation of Timber

- ❖ It makes timber durable
- ❖ It helps timber get rid of defects from fungi, insects, etc.
- ❖ If preservation of timber is done using oily preservatives, the glueing properties of timber increase.
- ❖ Preservation of timber using water-borne preservatives makes it fire-resistant to a large extent.



Demerits of Preservation of Timber

- ❖ Sometimes preservatives can leave an oily, unpaintable surface
- ❖ Some applications provide less physical protection to wood
- ❖ Preservation of timber leave a strong odour which is toxic and irritating to plants, animals and humans
- ❖ Dark-coloured timber is obtained when preservation of timber is done using oil preservatives.
- ❖ Oily, unpaintable surfaces are two major defects when preservation is of oily nature.
- ❖ Tendency to bleed or exude from the wood surface.

THANK
YOU